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# **FUNCTIONAL SPECIFICATION DOCUMENT (FSD)**

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# Micro Finance Limited

## 1. Business Objective

The objective of this system is to:

- Automate - **loan repayment collection processing**
- Handle - **partial payments and top-up loans**
- Ensure accurate - **interest calculation and outstanding balance tracking**
- Maintain proper - **audit trail (receipts/proof)**
- Support - **daily batch processing** for loan updates

## 2.Functional Requirements

**Customer Management:** Capture and manage customer details.

Customer Inputs:

- Name, Address, Aadhar
- Income, PAN, Age
- Phone, Nominee details, Loan required

Features:

- New customer registration
- Existing customer lookup
- Duplicate detection (PAN/Aadhar)

Business Rules:

- Customer must pass KYC validation
- Duplicate records should be flagged

**Loan Management:** Handle loan approval and lifecycle

Features:

- Loan request processing
- EMI plan selection (1/3/6/9 /12/15months)
- Daily EMI calculation
- Loan account creation

Business Rules:

Eligibility based on:

- Income
- Past history
- Rejected customers: Block for 3 months

Top-up loans allowed if:

- No overdue
- Good repayment history

**Payment processing:** Manage daily EMI collections.

Features:

- Daily due list
- Payment entry: Paid / Not Paid
- Advance payment handling
- Skip logic

Business Rules:

- Advance payment: Future days marked as paid
- Missed payment: Increase overdue count
- Early closure: Loan marked as closed, Removed from due list

**Inquiry Module:** Search, view and update customer/loan details.

Features:

Search by: Customer ID, Name

Output:

- Personal details
- Loan details
- Balance
- Overdue
- Missed payments
- Transaction history
- Loan status

**Alerts & Monitoring:** Track risky and overdue customers.

Features:

- Missed payments tracking
- Overdue alerts
- Risk classification

Alert Types:

- Missed > 3 days
- High overdue amount
- Frequent defaulters

Agent View: Only their village data

## Admin Module

Features:

- Create Mandals, Create Villages
- Assign Agents, Define rules (penalty, eligibility)

## Access Control

Role-Based Access:

Role Access

Admin Full access

Agent Only assigned village

Data Requirements

Core Entities:

- Customer
- Loan
- Payment
- Agent
- Alerts

Key Attributes:

Customer: Customer Id, Name, Address, Pan, Aadhar

Loan: Loan ID, Amount, EMI plan, Status

Payment: Date, Amount, Status

## Business Rules (Critical Section)

Loan Rules:

- EMI plans: 1 / 3 / 6 / 9 /12/15 months
- Daily EMI =  $[P + (P \times R \times M)] / (M \times 30)$

Where:

P = Loan amount

R = Monthly interest (5% = 0.05)

M = Number of months

Rejection Rule:

If rejected → cannot apply for 3 months

Payment Rules:

- Advance payment allowed
- Skip allowed if already paid

- Missed payment → overdue
- Fore closure

## Closure Rules:

If fully paid early:

- Mark closed
- Remove from active list

## Exception Handling

- Invalid KYC data
- Duplicate customer
- Payment mismatch
- Invalid loan request
- Foreclosure handling

## 3. In Scope / Out of Scope

### In Scope:

- Processing: Daily loan repayment transactions
- Handling:
  - ↳ Full payment
  - ↳ Partial payment
  - ↳ Top-up loan adjustment
  - ↳ Interest calculation (based on tenure)
- Updating:
  - ↳ Outstanding balance
  - ↳ Loan status
- Generating:
  - ↳ Updated loan file
  - ↳ Error /reject file

### Out of Scope:

- UI / Frontend screens
- Mobile Application
- Multi-Currency Support
- Advanced Interest Models
- No Multiple Active Loans
- No Real Time Notifications
- No External Credit Investigations
- No Sureties

### Future Enhancements:

- Customer Portal
- Online Payments

- Multi Active Loans

## 4. Assumptions

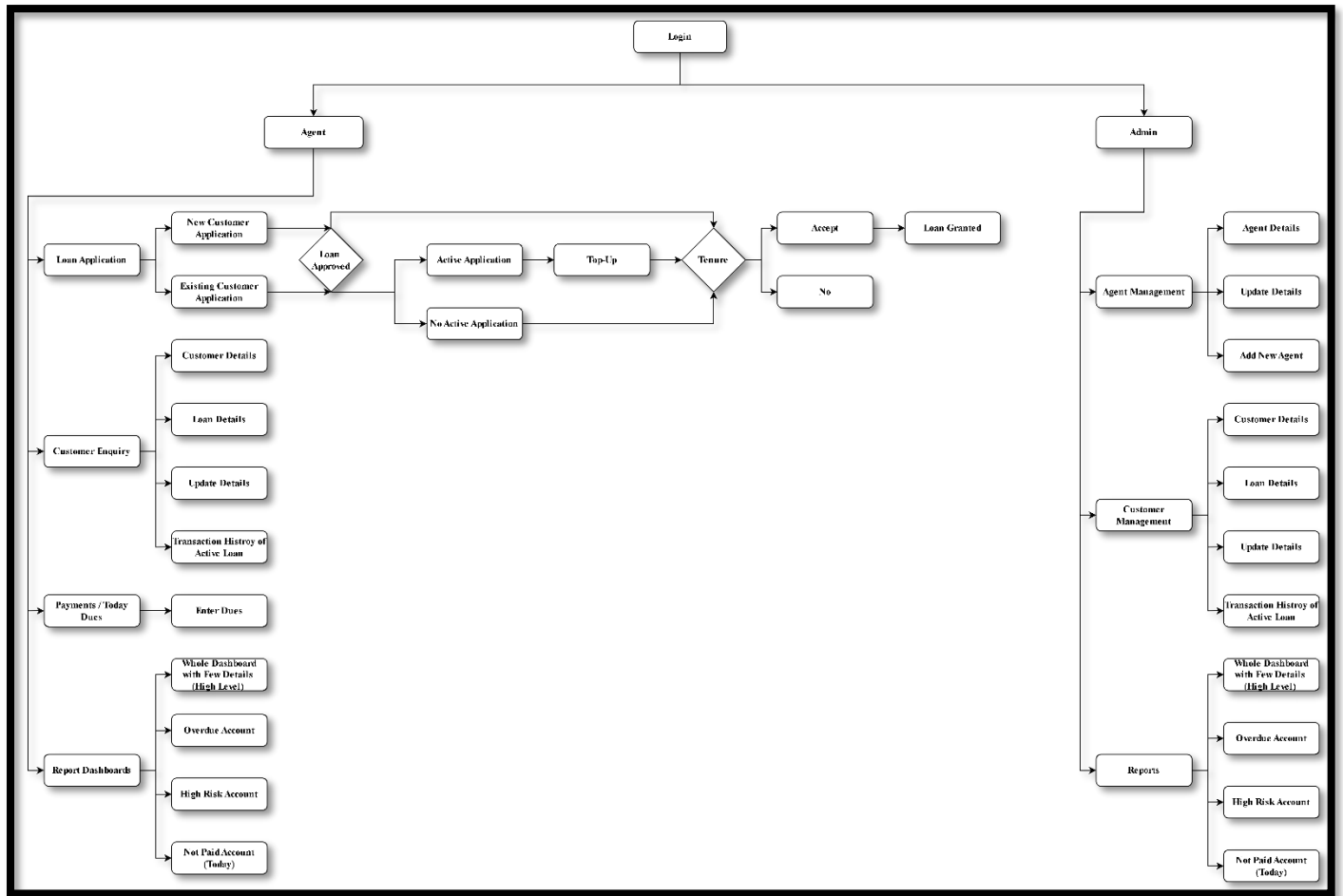
- Each month = **30 days**
- Interest rate is **fixed per loan**
- Input file is **pre-validated and sorted**
- Customer has only **one active loan at a time**
- All transactions are processed in **batch mode (EOD)**
- Currency = INR (no multi-currency handling)

## 5. High-Level Flow

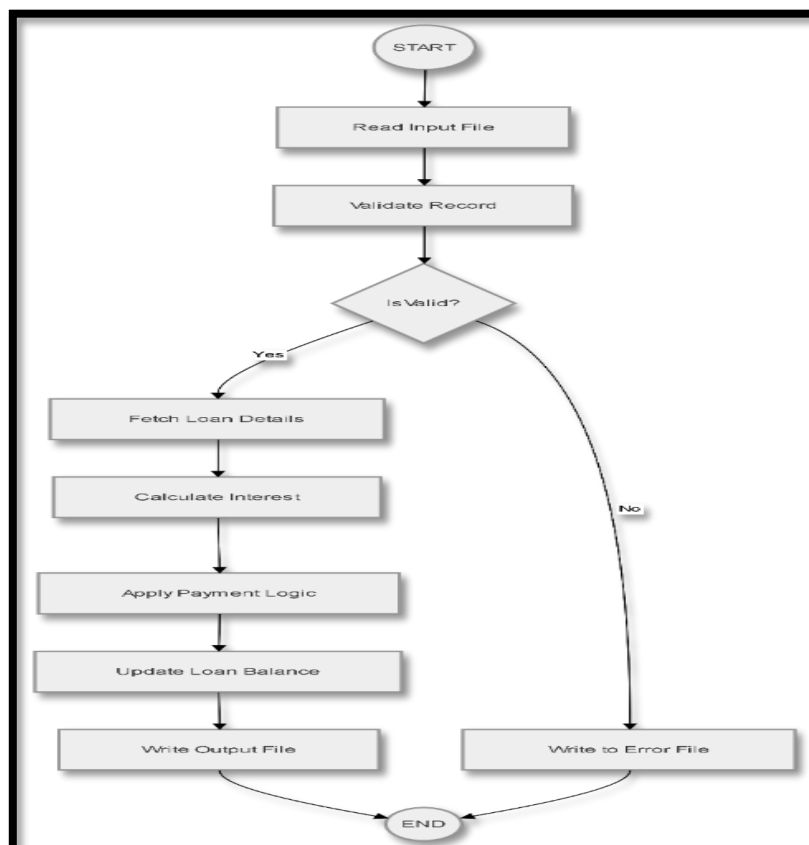
Simple understanding of system flow:

1. Read Input Loan Repayment File
2. Validate data
3. Fetch existing loan details
4. Calculate:
  - ↳ Interest
  - ↳ Remaining balance
5. Apply payment:
  - ↳ Full /Partial /Top-up logic
6. Update loan record
7. Write:
  - ↳ Updated loan file
  - ↳ Error file (if any)

## 6. Overall Flow (Flowchart)



### Flow Explanation:



## 7. Sample Screens

MICRO FINANCE LIMITED

Date: 08/04/2026Time: 10:15:32

USER ID : AGT101

PASSWORD : \*\*\*\*\*

[ENTER] Login

PF3: Exit PF5: Clear PF9: Home PF12: Logout

MESSAGE:

Login Successful

### When Agent Selected Option 1

MICRO FINANCE LIMITED

Date: 08/04/2026Time: 10:16:10

AGENT MENU

1. Loan Application

2. Customer Enquiry

3. Today Dues / Payments

4. Dashboard

Select Option: 1

PF3: Exit PF9: Home PF12: Logout

MESSAGE:

Navigating to Today Dues...

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## LOAN APPLICATION

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1. New Customer

2. Existing Customer

Select Option: \_

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[ENTER] Continue

PF3: Back    PF9: Home

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MESSAGE:

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## NEW CUSTOMER APPLICATION

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Name : \_\_\_\_\_

Address : \_\_\_\_\_

Age : \_\_\_\_

Phone Number : \_\_\_\_\_

Aadhar Number : \_\_\_\_\_

Profession : \_\_\_\_\_

Monthly Income : \_\_\_\_\_

Loan Amount Req : \_\_\_\_\_

Father/Spouse Name : \_\_\_\_\_

Father Phone : \_\_\_\_\_

Nominee Name : \_\_\_\_\_

Nominee Phone : \_\_\_\_\_

Relation : \_\_\_\_\_

-----  
[ENTER] Submit

PF3: Back PF5: Clear PF9: Home

-----  
MESSAGE:  
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-----  
EXISTING CUSTOMER APPLICATION  
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Customer Name : \_\_\_\_\_

Phone Number : \_\_\_\_\_

Loan Amount Request : \_\_\_\_\_

-----  
[ENTER] Fetch Details

PF3: Back PF9: Home

-----  
MESSAGE:  
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## LOAN ADJUSTMENT (TOP-UP)

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Customer ID : \_\_\_\_\_

Name : \_\_\_\_\_

Previous Loan Amt : \_\_\_\_\_

Amount Paid : \_\_\_\_\_

Outstanding : \_\_\_\_\_

New Loan Requested : \_\_\_\_\_

Eligible Amount : \_\_\_\_\_

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Adjustment:

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Old Loan Closure : \_\_\_\_\_

Cash to Customer : \_\_\_\_\_

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[ENTER] Approve & Create Loan

PF3: Back    PF9: Home

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MESSAGE:

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## When Agent Selected Option 2

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### CUSTOMER ENQUIRY

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Customer ID : \_\_\_\_\_

Customer Name : \_\_\_\_\_

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[ENTER] Fetch

PF3: Back PF9: Home

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MESSAGE:

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### CUSTOMER DETAILS

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Customer ID : \_\_\_\_\_

Name : \_\_\_\_\_

Phone : \_\_\_\_\_

Address : \_\_\_\_\_

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Loan Summary

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Active Loan ID : \_\_\_\_\_

Loan Amount : \_\_\_\_\_

Outstanding : \_\_\_\_\_

Daily EMI : \_\_\_\_\_

DPD : \_\_\_\_\_

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PF5: Update Details   PF6: Transaction History

PF3: Back              PF9: Home

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MESSAGE:

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## TRANSACTION HISTORY

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Customer ID : \_\_\_\_\_

Filter:

1. Last Week

2. Last Month

3. All

4. Date Range

Payment Filter:

1. All

2. Paid

3. Not Paid

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| Date | Due | Paid | Status | Reason |
|------|-----|------|--------|--------|
|------|-----|------|--------|--------|

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01-APR |100 |100 | PAID | -

02-APR |100 |0 | NOT PD | Not Available

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PF3: Back PF9: Home

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### When Agent Selected Option 3

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#### TODAY DUES / PAYMENTS

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Date: \_\_/\_\_/\_\_\_\_ Agent ID: \_\_\_\_\_

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| CustID | Name | LoanID | Due | Pending | Paid Amt | Status | Reason |
|--------|------|--------|-----|---------|----------|--------|--------|
|--------|------|--------|-----|---------|----------|--------|--------|

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|      |     |      |     |     |       |   |       |
|------|-----|------|-----|-----|-------|---|-------|
| 1001 | Ram | L001 | 100 | 500 | _____ | _ | _____ |
|------|-----|------|-----|-----|-------|---|-------|

|      |     |      |     |      |       |   |       |
|------|-----|------|-----|------|-------|---|-------|
| 1002 | Raj | L002 | 100 | 1200 | _____ | _ | _____ |
|------|-----|------|-----|------|-------|---|-------|

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Status:

P = Paid N = Not Paid

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[ENTER] Submit Payments

PF3: Back PF5: Refresh PF9: Home

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MESSAGE:

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## When Agent Selected Option 4

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### DASHBOARD

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Date: \_\_/\_\_/\_\_\_\_ Agent ID: \_\_\_\_\_

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### ALERTS

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High Risk Accounts : \_\_ → PF1

Not Paid Today : \_\_ → PF2

Overdue Accounts : \_\_ → PF4

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### TODAY SUMMARY

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New Accounts : \_\_\_\_\_

Settled Accounts : \_\_\_\_\_

Collected Amount : \_\_\_\_\_

Pending Amount : \_\_\_\_\_

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### PERFORMANCE

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Target Amount : \_\_\_\_\_

Achievement % : \_\_\_\_\_

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PF1: High Risk PF2: Not Paid PF4: Overdue

## 8. Detailed Process Logic

### Step 1: Input Processing

- Read each record:
  - ↳ Customer ID
  - ↳ Loan Amount
  - ↳ Payment Amount
  - ↳ Tenure

### Step 2: Validation

- Customer exists
- Loan is active
- Payment amount > 0

### Step 3: Interest Calculation

Formula:  $\text{Interest} = (\text{Loan Amount} \times \text{Rate} \times \text{Days}) / 100$

Where:

$$\text{Days} = \text{Months} \times 30$$

### Step 4: Payment Handling

- ↳ Case 1: Full payment
  - Loan closed
  - Balance=0
- ↳ Case 2: Partial payment
  - Deduct from outstanding
  - Keep loan active
- ↳ Case 3: Top-Up Loan
  - If previous loan partially paid then,
  - New loan sanctioned = Remaining eligibility
  - Old loan adjusted/closed

## Step 5: Output Generation

↳ Success file

- Update loan details

↳ Error file

- Invalid records with reason

## 9. File attributes

Required files are:

↳ MASTER FILE

↳ TRANSACTION FILE

↳ AGENTFILE

↳ WORK FILE

↳ AUDIT FILE

### 9.1 Master file

01 MASTER-RECORD.

05 LOAN-KEY.

10 LOAN-ID PIC X (10).

10 CUSTOMER-ID PIC 9(6).

05 CUSTOMER-DETAILS.

10 CUSTOMER-NAME PIC X (30).

10 PHONE-NUMBER PIC 9(10).

10 AADHAR-NUMBER PIC 9(12).

10 PAN-NUMBER PIC X (10).

10 ADDRESS PIC X (50).

10 INCOME PIC 9(8)V99.

05 LOAN-DETAILS.

10 LOAN-AMOUNT PIC 9(9)V99.

10 INTEREST-RATE PIC 9(2)V99.

10 TENURE-MONTHS PIC 9(2).

10 START-DATE PIC 9(8).

10 END-DATE PIC 9(8).

10 EMI-AMOUNT PIC 9(7)V99.  
10 LOAN-STATUS PIC X (1).  
05 PAYMENT-SUMMARY.  
10 TOTAL-PAID PIC 9(9)V99.  
10 OUTSTANDING PIC 9(9)V99.  
10 LAST-PAYMENT-DATE PIC 9(8).  
05 COLLECTION-TRACKING.  
10 PAID-DAYS PIC 9(3).  
10 MISSED-DAYS PIC 9(3).  
10 OVERDUE-DAYS PIC 9(3).  
05 AGENT-LINK.  
10 AGENT-ID PIC X (6).  
05 STATUS-FLAGS.  
10 DEFAULTER-FLAG PIC X (1).  
10 ALERT-FLAG PIC X (1).

## 9.2 Transaction file

01 TRANSACTION-RECORD.  
05 TRANSACTION-ID PIC 9(10).  
05 KEYS.  
10 CUSTOMER-ID PIC 9(6).  
10 LOAN-ID PIC X (10).  
10 AGENT-ID PIC X (6).  
05 CUSTOMER-INFO.  
10 CUSTOMER-NAME PIC X (30).  
05 TRANSACTION-DATA.  
10 TRANSACTION-DATE PIC 9(8).  
10 TRANSACTION-TYPE PIC X (2).  
05 PAYMENT-DETAILS.  
10 EMI-DUE-AMOUNT PIC 9(7)V99.  
10 AMOUNT-PAID PIC 9(7)V99.

05 STATUS-DETAILS.

10 PAYMENT-STATUS PIC X (1).

10 OVERDUE-AMOUNT PIC 9(9)V99.

10 MISSED-DAYS PIC 9(3).

10 REASON PIC X (50).

### 9.3 Agent file

01 AGENT-RECORD.

05 AGENT-ID PIC X (6).

05 AGENT-NAME PIC X (30).

05 VILLAGE-CODE PIC X (6).

05 STATUS PIC X (1).

05 NO OF CUSTOMERS

05 NO OF OVERDUE CUSTOMERS